

2 Egg cells vary in size.

(a) The table shows the size of egg cells from three species of animal.

Species	Diameter of egg cell / μm
Orb weaver spider	10
Human	150
Blue poison dart frog	1600

- (i) Which of the following shows the diameter, in standard form, of the egg cell of a blue poison dart frog?

(1)

- ☐ A $1.6 \times 10^{-2} \mu\text{m}$
- ☐ B $1.6 \times 10^2 \mu\text{m}$
- ☐ C $1.6 \times 10^{-3} \mu\text{m}$
- ☐ D $1.6 \times 10^3 \mu\text{m}$

- (ii) The egg cells of humans and blue poison dart frogs are approximately spherical.

The volume of a sphere (V) is calculated using the formula

$$V = \frac{4}{3}\pi r^3$$

The volume of the human egg cell is $1.8 \times 10^6 \mu\text{m}^3$.

Calculate how many times larger the volume of the blue poison dart frog egg cell is than the volume of a human egg cell.

Give your answer to two significant figures.

(3)

Answer



(b) An egg cell and a skin cell were removed from a female human.

Both cells contained a nucleus.

The skin cell contained 46 chromosomes.

Explain the differences between the genetic material from these two cells.

(4)

(Total for Question 2 = 8 marks)

